BOSCH 博世

Confidential

Product Development Design Change Request
产品开发设计变更请求

Version 15

Effective date: 2025-12-01th

Customer project Name/ 客户项目名称:	JIM-4JJ	MCR No. / MCR号:	
Design Change Request No.	25_128		
Sample status / 样件状态:	☐ A Sample ☐ B Sample ☐ C Sample ☑ D Sample		

Change part&product Name/更改零部件产品的名称(简要描述)

DOC DPF :F01ZH003EA-00
SCR: F01ZH003EB-00
DPF BRACKET:F01Z5017P5-05

Reason of changes/更改理由 (简要描述):

DPF 出气法兰，客户装上双头螺柱后，有密封垫片，安装不上的风险，需要增加 对出气法兰螺母的位置检测
SCR 客户要求 在尾管处增加纸质标签
侧支架 客户要求 更改客户料号

Change from/变更来源:

Customer request / 客户要求 ☑	Supplier request / 供应商请求 ☐	Design optimization / 设计优化 ☐
Correction / 更正(错误) ☐	Other / 其他 ☐ ( )	

Change inform to/变更通知人: Zhang Shengchang,Zhang Hanquan,Shen Weibo，Li Yicen，Fang Jiayan，Yin Kangzhuang Yang Guangtuo,Jia Kebin，Luo jing，Lu Qingsong,Xie Youfu

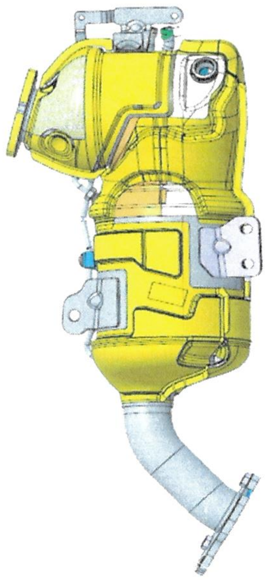
Whether this change impact others component or product 此变更是否影响到其他部件和产品: ☐ yes/是 ☑ no/否

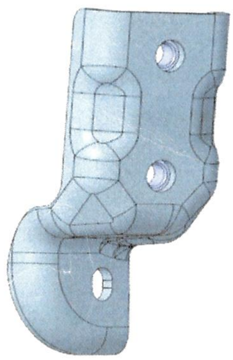
If impact others product，Whether others projects&product change or not? 如果影响，别的部件和产品是否也需要同时变更?

Design / 设计工程师 Signature / 签名:	孔坤阳	Date / 日期:	2025.12.01
Check / 校对工程师 Signature / 签名:	杨广拓	Date / 日期:	2025.12.02

Description of Change/更改描述:（下面空白处请列出详细的变更前后对比)

基于C样件模型未更改
详细更改点见附件





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Design Change Review Sheet
设计变更评审表

Version 13
Effective date: 2025-09-25th

Design Change Request No.

1. Function&Performance will be influenced? / 系统性能影响? ☐ yes/是 ☑ no/否		
If Yes, System impaction evaluation? 如果是，系统影响评估? (如果是初次样件放行，空白处请说明样件目的)	评估对系统无影响	System/签名 [illegible]
Whether need sample to test? Plan? 是否需要样件进行验证? 计划?	NA	
Whether the catalyst information is confirmed? 催化剂信息是否确认? (物料号、图纸、SPEC)		Catalyst/签名 [illegible]
2. Interface&boundary (internal and customer) will be influenced? / 接口&边界(内部和客户)影响? ☐ yes/是 ☑ no/否		
If impact custmer ,whether get confirm from customer side? 请说明影响和是否得到客户的确认?	模型无更改	Design /签名 马坤阳
3. Mechanical strength&Durability will be influenced? / 机械强度&可靠性影响 ? ☐ yes/是 ☑ no/否		
If Yes, How to evaluate? Plan? 如果是，请说明影响和如何评估,计划? (例如:仿真分析等)		
Whether need sample to test? Plan? 是否需要样件进行验证? 计划?	设计无变更	Validation/签名 [illegible]
4. Product document influenced?		
4.1 Interface FMEA-relevant / IFMEA:	☐ no/否 ☐ yes/是 Note____	
4.2. Product FMEA-relevant / DFMEA:	☐ no/否 ☐ yes/是 Note____	
4.3. Special Charecteristics /PSC: (针对C Sample)	☐ no/否 ☐ yes/是 Note____ OK	
4.4. IMDS relevant: (针对C Sample)	☐ no/否 ☐ yes/是 Note____	
4.5 Offer drawing relevant:(针对C Sample)	☐ no/否 ☐ yes/是 Note____	
4.6. TCD relevant:	☐ no/否 ☐ yes/是 Note____	Design /签名 马坤阳
5. Changed parts Stock& treatment / 需要变更零件的库存统计和处理? (只针对涉及到已经生产过的零件变更，新放行零件和本条不相关) ☐ NA.(新零件放行,不涉及库存) ☐ 涉及到已经生产过的零件变更		
RBCW RDC(外库)和RBCW 仓库，是否有库存，	☐ 无库存 ☐ 有库存, 库存数量____	
供应商处(已做好，未发货)是否有库存，	☐ 无库存 ☐ 有库存, 库存数量____	
LPN工厂是否有库存，	☐ 无库存 ☐ 有库存, 库存数量____	
在途，运输中是否有库存	☐ 无库存 ☐ 有库存, 库存数量____	PUE /签名 [illegible]
被影响到的实物库存处理指示: (设计工程师组织会议讨论)	☑ 可以继续使用完 ☐ 改制后使用 ☐ 报废	Design /签名 马坤阳
	☐ 制费用____; 报废成☐____;	PUE /签名____
已装车件处理方法 (发客户或标定车等) (SMP组织会议讨论)	☐ 制____; 更换☐____; 其他 (单独说明);	
6. Influence on supplier part?		
Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier? 变更后的零件，生产可行性供应商是否确认? 反馈?	邮件 零件无变更，按图纸新零件采购.	
Influence on purchasing part delivery time? 对采购件交货时间的影响?		
涉及到变更的零件，博世已出PR，供应商还未来得及生产的订单，是否已经通知供应商取消.	☐ no/否 ☐ yes/是	
Does the change influence on purchasing cost ? 变更对采购成本的影响?	☐ increase/增加,金额____ ☐ decrease/降低 ,金额____ ☐ no change/不变	PUE/签名 [illegible]
7. DFMA & Sample prodcution OPL?		
必要时附加DFMA 文件	无OPL	Design /签名 马坤阳
8. Influence on LPN manufacturing&assembly process?		
Feedback from manufacturing engineer? 生产制造装配是否影响，反馈?		
Influence on project&product delivery time? 对项目&产品交货时间的影响?	PPC 沿用（[illegible]	
Influence on manufacturing cost / 对生产成本的影响?	☐ increase/增加 ,金额____ ☐ decrease/降低 ,金额____ ☐ no change/不变	PUE/签名 [illegible]
9. PM alignment? (Influence on project Timeline? Cost?); 是否同意对项目时间和成本的影响____; 需求样品订单数量____;		PM/签名 [illegible]
10. 如果涉及以下情况需TCR3 审核确认。 不涉及则 NA.: ☐ ☐ 库存报废,金额____; ☐ 采购成本增加,金额____; 生☐成本增加,金额____		RBCW/TCR3

Design Team Leader	ENG-CV1	PUR	TCR3
Signature / 签名: [illegible]	Signature / 签名: [illegible]	Signature / 签名: [illegible] 2025.12.4	Signature / 签名: [illegible] 2025-12-03

ATS RACE Workflow

ATS RACE release Checklist(C to D sample drawing release)

	Release Points	Result/Status	Deadline	Resp.
Design	1. Final 3D model and offer drawing officially confirmed by customer?			
	2. Final nameplate confirmed by customer?	OK		[illegible] 2025.12.02
	3. IMDS or CAMDS finished?			
	4. D-FMEA closed and signed ? Sign?			
	5. I-FMEA signed with customer? Sign?	coveed by 493		
	6. Installation instruction finished?			
	7. Finial SC/CC finished by EGT5/MOE/QMM/PUR? Sign?	covered by PHA.		
MOE	1. Final welding process flow chart finished by MOE?	done		
	2. Welding and assembly trial run finished?	ok		[illegible]
	3. P-FMEA signed ?			
Validation	1. 9000 Km bad road Vehicle validation status?			
	2. PV report?	OK (见附件)		
System	1. Customer validation report? (If customer take responsible the calibration, take as reference)	Covered by 9000km		[illegible]
	2. TCD is available?	OK		
	3. PD is available?	OK		[illegible]
Flow	1. QG0 and QG1 passed? QGC2 plan?	OK N/A		[illegible]

**: Finished w/o open points; RACE can be proceeded.

Finished w/ open points and countermeasures are available. Escalation to SM and inform to PM for RAC release or not.

Finished w/ open points but root cause is not clear and countermeasure is not available; Inform PM and reject the RACE with inputting the reason in reject step.

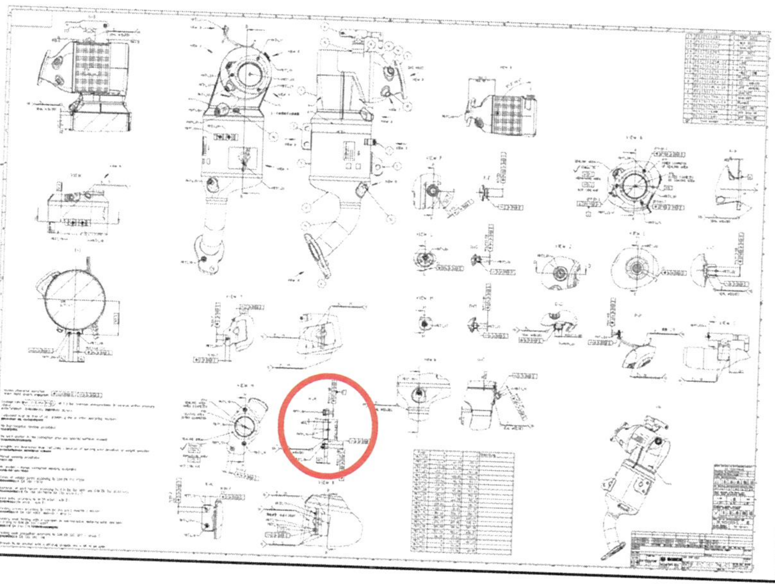
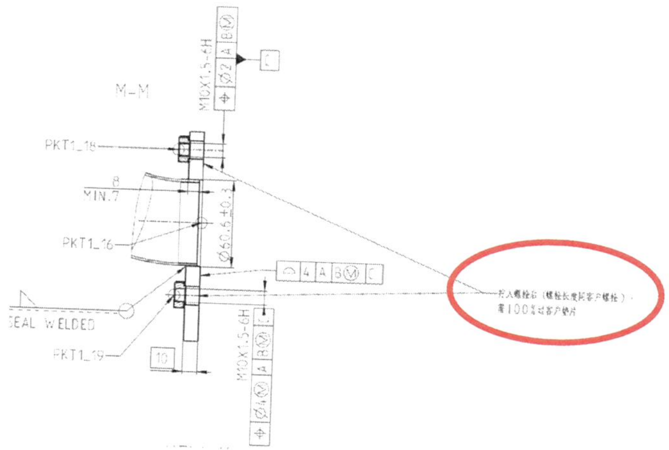
***: Design SPjM responsible for the checklist document, but relative points should be checked with the Resp. person.

PDECR25-128（4JJ1-D样件）

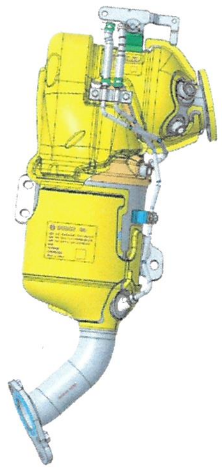
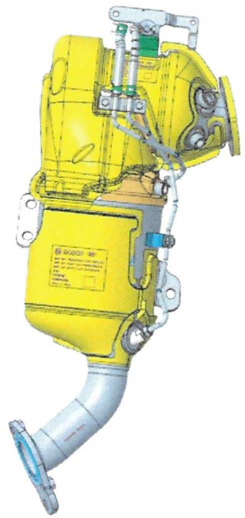
ENG-CV

BOSCH

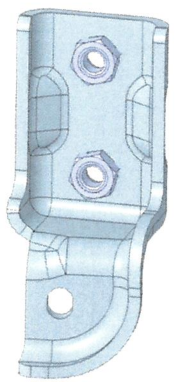
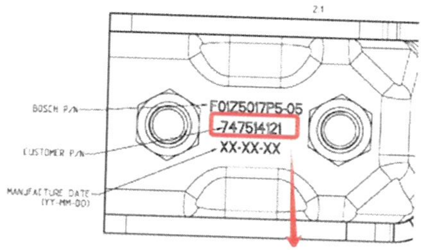
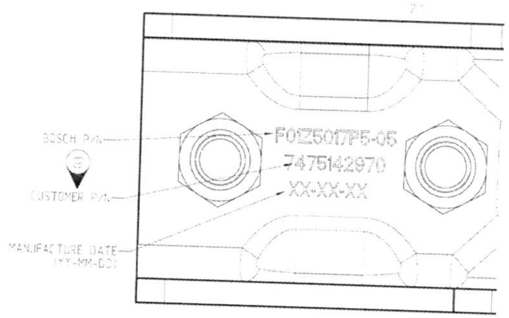
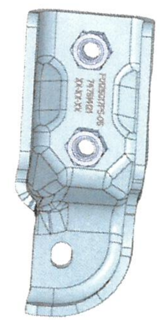
PDECR 25-128

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017NL-05	DOC+SDPF WELDING	JIE-4JJ	D	BOSCH/Customer
变更描述	变更前 F01Z5017NL-03 		变更后  增加新的检测方法: 拧入螺栓后（螺栓长度同客户螺栓）， 需100%过客户垫片		
变更原因	增加新的检测方法;				
风险点（影响）	无				
验证方法	无				

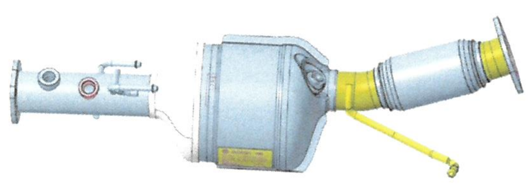
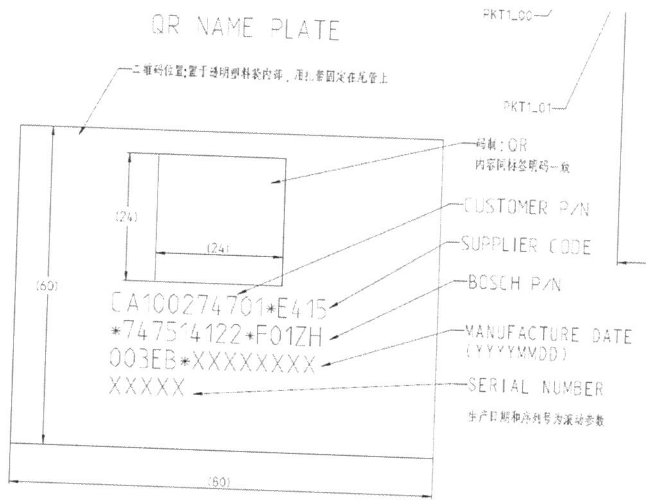
PDECR 25-128

	零件号	零件名称	Project	样件阶段	变更来源
	F01ZH003EA-00	DOC+SDPF UNIT	JIE-4JJ	D	BOSCH/Customer
变更描述	变更前		变更后		
				料号更改，模型无变更 由F01ZH003EA-03改为F01ZH003EA-00	
变更原因	料号更改				
风险点（影响）	无				
验证方法	无				

PDECR 25-128

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017P5-05	BRACKET	Jle-4JJ	D	BOSCH/Customer
变更描述	变更前 F01Z5017P5-05 		变更后  		 客户料号更新为 7475142970
变更原因	客户要求				
风险点（影响）	无				
验证方法	无				

PDECR 25-128

	零件号	零件名称	Project	样件阶段	变更来源
	F01ZH003EB-00	SCR	Jle-4JJ	D	BOSCH/Customer
变更描述	变更前 F01ZH003EB-05 		变更后   此次图纸更新件号升级为 CA100274701			
变更原因	SCR 客户要求 在尾管处增加纸质标签				
风险点（影响）	无				
验证方法	无				

PDECR 25-128

零件号	零件名称	Project	样件阶段	变更来源
F01Z5102T9-00	DPF MAT	Jle-4JJ	C	BOSCH/Customer

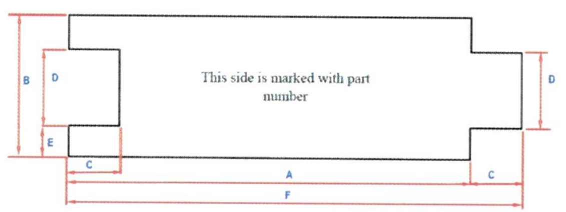
变更描述

变更前

UNIFRAX 奇耐联合纤维（上海）有限公司

Customer Name/客户名称	Material Type/材料类型 (FBW)		BBW	膜重	Part No./零件号	Date/日期
Bosch	CC-Max 5C	1800	1800	0	4JJ 172X177.8 DPF	2025.9.2

No Scrim / Blue Scrim / Green Scrim / Rough Side 粗糙面 / Smooth Side 光滑面



特殊要求/Special Requirements:

Dimension/尺寸	A[mm]	B[mm]	C[mm]	D[mm]	E[mm]	F[mm]	Thickness[mm]	Basis Weight[g/m2]	Part Weight[g]
Normal/标准	578.0	163.0	40.0	53.0	55.0	618	13.50	1800	169.59
Tolerance/公差	2.5		1	1	1				
Max/最大值	580.5	164.0	41.0	54.0	56.0		14.9	1944	186.54
Min/最小值	575.5	162.0	39.0	52.0	54.0		12.1	1656	152.63

制 图 / 日期 | 客户确认/ 日期

备注/Remark

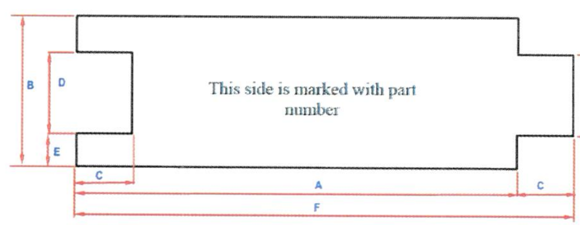
变更后

UNIFRAX 奇耐联合纤维（上海）有限公司

Customer Name/客户名称	Material Type/材料类型 (FBW)		BBW	膜重	Part No./零件号	Date/日期
Bosch	CC-Max 5C	1800	1800	0	4JJ 172X177.8 DPF	2025/10/23

Prototype / OTS / PPAP / SOP

No Scrim / Blue Scrim / Green Scrim / Rough Side 粗糙面 / Smooth Side 光滑面



特殊要求/Special Requirements:
1. 喷涂零件码

Dimension/尺寸	A[mm]	B[mm]	C[mm]	D[mm]	E[mm]	F[mm]	Thickness[mm]	Basis Weight[g/m2]	Part Weight[g]
Normal/标准	578.0	158.0	40.0	53.0	55.0	618	13.50	1800	164.38
Tolerance/公差	2.5		1	1	1				
Max/最大值	580.5	159.0	41.0	54.0	56.0		14.9	1944	180.82
Min/最小值	575.5	157.0	39.0	52.0	54.0		12.1	1656	147.94

制 图 / 日期 | 客户确认/ 日期

备注/Remark

变更原因	工厂反馈衬垫封装后，会超出催化剂，建议剪短5mm。经奇耐确认剪短5mm后，安全系数为1.1，评估风险较低，可以剪短
风险点（影响）	无
验证方法	无

Powertrain Solutions

BOSCH

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CC:

Tere25_029

Test Report Vehicle endurance test parts investigation

1. General information

Project name	JIE 4JJ ATS CN6	**MCR No.**	-
ATS concept	☒ in-line ☐ Box ☐ U-shape	**Test ground**	XIANGYANG
Part number	DOC+DPF: F01ZH003EA-02 SCR: F01ZH003EB-02	**ATS sample phase**	B sample

2. Operation condition

Test item	Comprehensive roads endurance test
Test description	As the design validation requirement for JIE 4JJ ATS CN6 project, 42000km comprehensive roads endurance test was performed to the ATS parts. The test purpose is to verify the resistance to the rough road load for ATS mechanical parts.

3. Test result

1) Sealing check

   The leakage rate of the ATS after test (DOC+DPF:0.157L/min, SCR:0.035L/min) can fulfill the design requirement (DOC+DPF:<0.5L/min, SCR:<0.3L/min @filling pressure 30±5kpa).

2) Visual inspection

   All components visual check are in acceptable condition.

4. Conclusion

☒ **Pass** ☐ **Pass with limitation** ☐ **No pass**

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10.13.2025

5. Investigation

1) Leakage test for the whole ATS

- After the endurance test, the leakage test was performed to the DOC+DPF and SCR units. From the leakage test results (table 1-2), the leakage rate of the ATS after test can fulfill the design requirement.

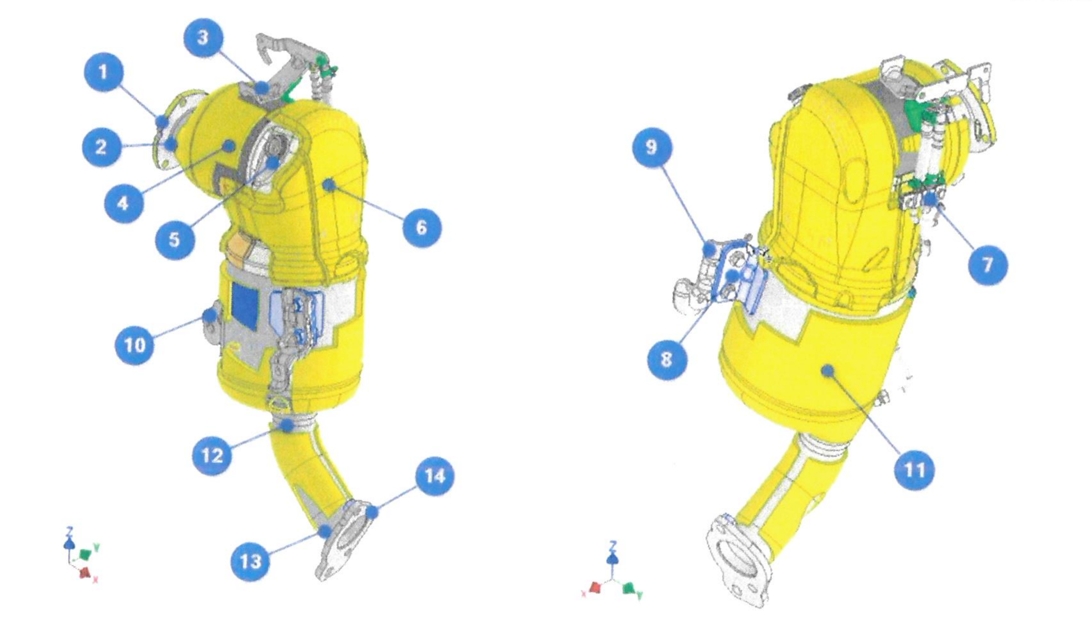
Table 1 Leakage test result for DOC+DPF

Leakage rate result before and after test @30kPa filling pressure		
	After testing	Test requiremnt
Leakge rate (L/min)	0.157	<0.5

Table 2 Leakage test result for SCR

Leakage rate result before and after test @30kPa filling pressure		
	After testing	Test requiremnt
Leakge rate (L/min)	0.035	<0.3

2) DOC+DPF investigation after test



Part No.	components	No failure	Failure	Figure
1	DOC inlet flange	×		Fig.3
2	DOC inlet flange welding seam	×		Fig.4-5
3	ΔP sensor bracket and welding seams	×		Fig.6-7
4	DOC catalyst	×		Fig.9-10

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10.13.2025

5	DM flange	×		Fig.21-22
6	Mixer plate	×		Fig.11-12
7	ΔP pipe bracket and welding seams	×		Fig.8
8	DPF left mounting bracket 1	×		Fig.13-14
9	DPF left mounting bracket 2	-		-
10	DPF right mounting bracket	×		Fig.15-16
11	DPF catalyst	×		Fig.17
12	DOC outlet flange welding seam	×		Fig.18-19
13	DOC outlet flange	×		Fig.20

Fig.1 DOC+DPF overview

Fig.2 DOC+DPF overview

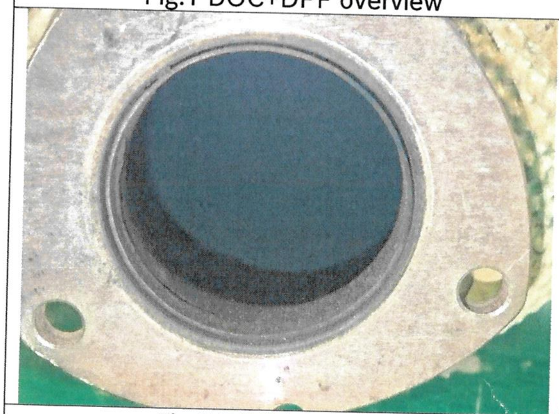

Fig.3 DOC inlet flange

Fig.4 DOC inlet flange welding seam

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10.13.2025

Tere25_029

Fig.5 DOC inlet flange welding seam

Fig.6 ΔP sensor bracket and welding seams

Fig.7 ΔP sensor bracket and welding seams

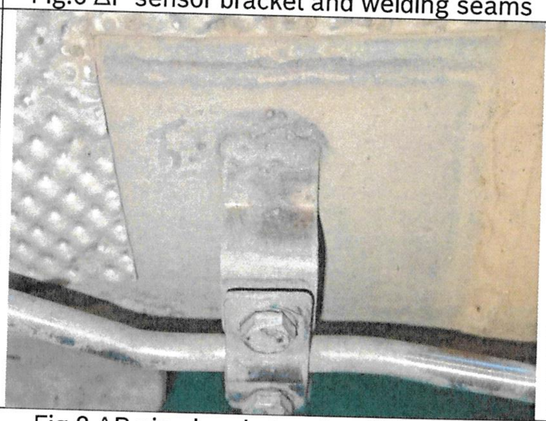

Fig.8 ΔP pipe bracket and welding seams

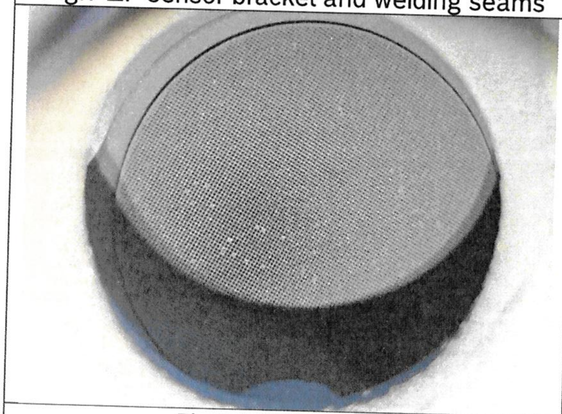

Fig.9 DOC catalyst

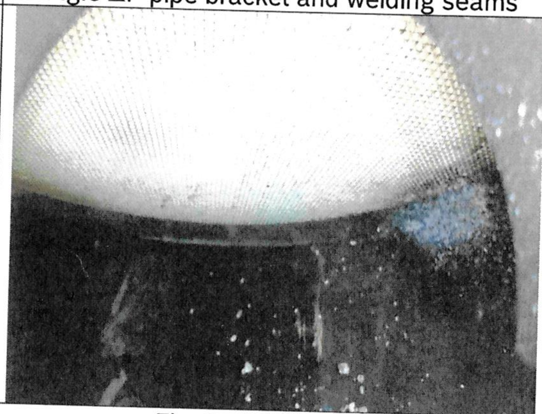

Fig.10 DOC catalyst

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10.13.2025

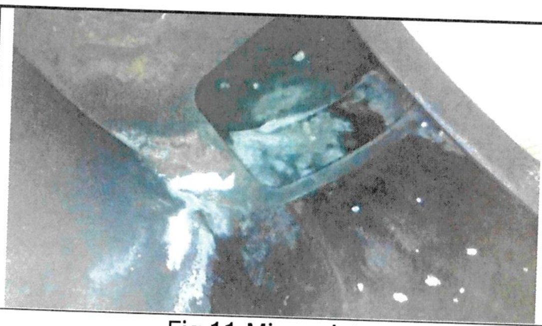
Fig.11 Mixer plate

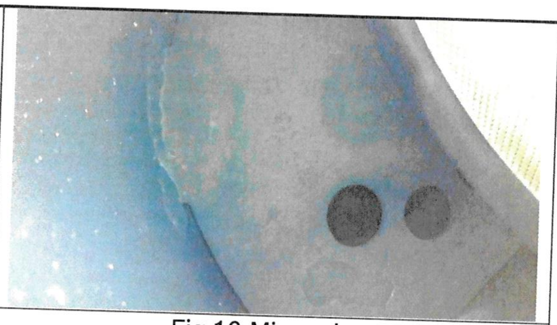
Fig.12 Mixer plate


Fig.13 DPF left mounting bracket 1


Fig.14 DPF left mounting bracket 1


Fig.15 DPF right mounting bracket

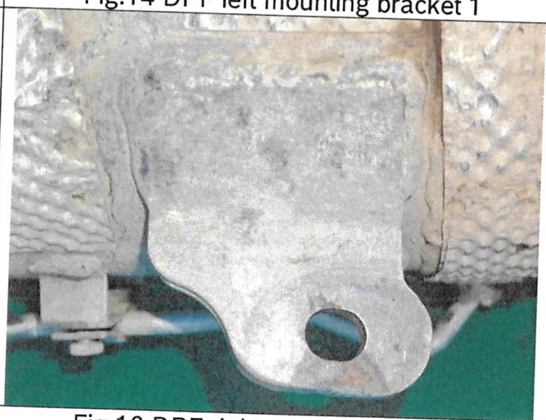
Fig.16 DPF right mounting bracket

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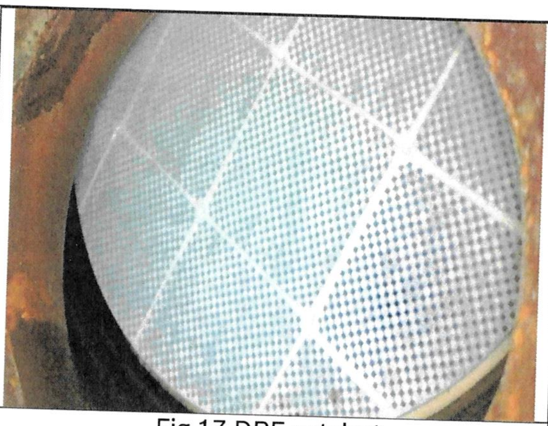

Fig.17 DPF catalyst

Fig.18 DPF outlet flange welding seam

Fig.19 DPF outlet flange welding seam

Fig.20 DPF outlet flange welding seam

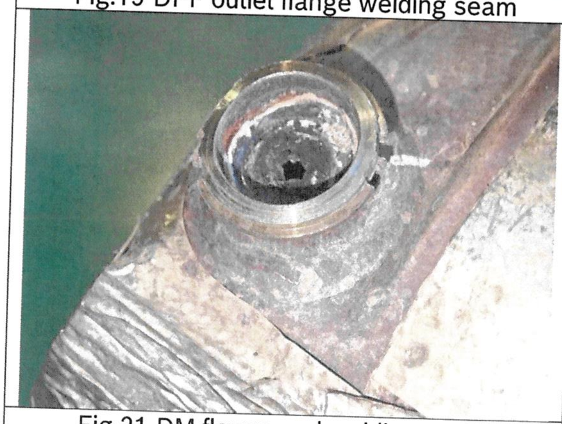

Fig.21 DM flange and welding seam

Fig.22 DM flange and welding seam

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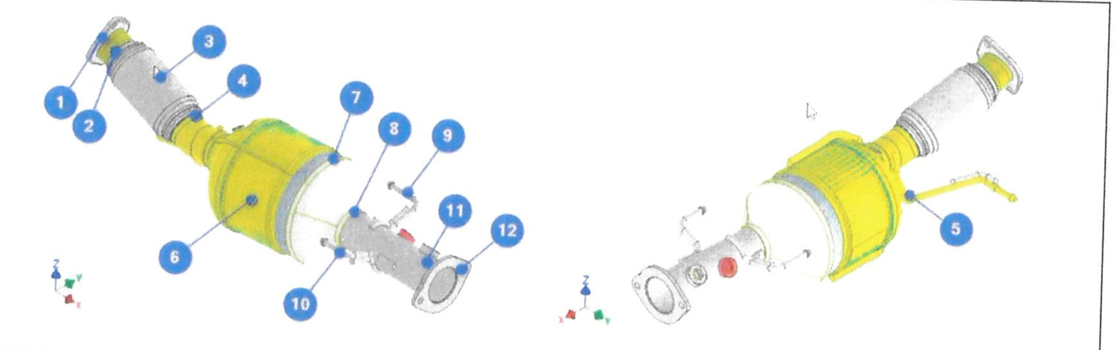
3) SCR investigation after test



Part No.	components	No failure	Failure	Figure
1	SCR inlet flange and welding seams	×		Fig.25-27
2	Flexible inlet welding seam	×		Fig.28-29
3	Flexible pipe	×		Fig.28-29
4	Flexible pipe outlet welding seam	×		Fig.28-29
5	Front hook and welding seam	×		Fig.30-32
6	SCR catalyst	×		Fig.33-34
7	SCR sleeve welding seam	×		Fig.23-24
8	SCR outlet pipe welding seam	×		Fig.35-36
9	Right rear hook and welding seam	×		Fig.37-38
10	Left rear hook and welding seam	×		Fig.39-40
11	SCR outlet flange welding seam	×		Fig.41-42
12	SCR outlet flange	×		Fig.43

Fig.23 SCR overview

Fig.24 SCR inlet flange

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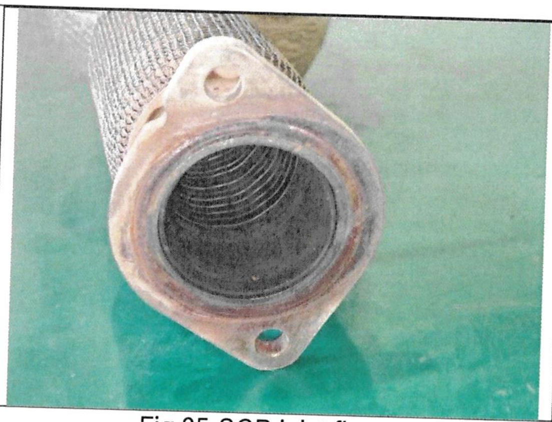

Fig.25 SCR inlet flange

Fig.26 SCR inlet flange welding seam

Fig.27 SCR inlet flange welding seam

Fig.28 Flexible pipe outlet welding seam

Fig.29 Flexible pipe outlet welding seam

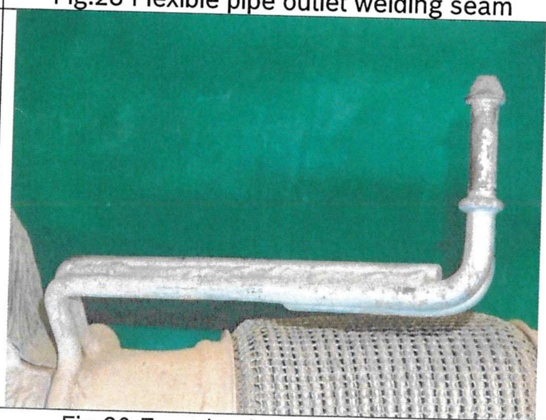

Fig.30 Front hook and welding seam

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Fig.31 Front hook and welding seam

Fig.32 Front hook and welding seam

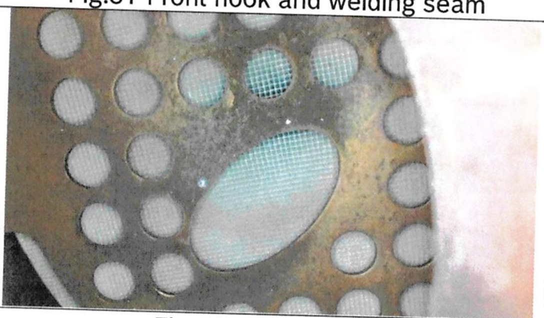

Fig.33 SCR catalyst

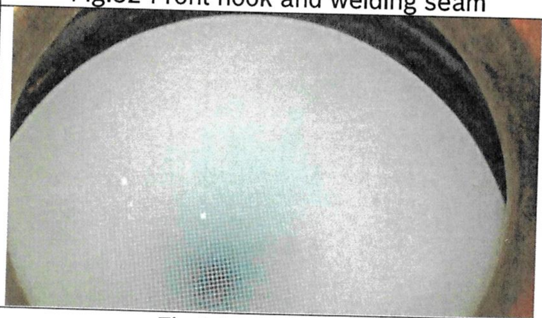

Fig.34 SCR catalyst

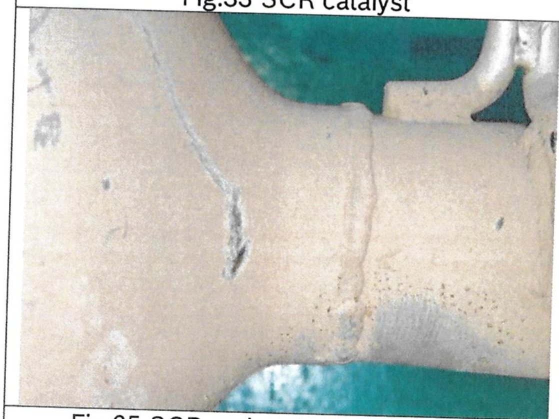

Fig.35 SCR outlet pipe welding seam

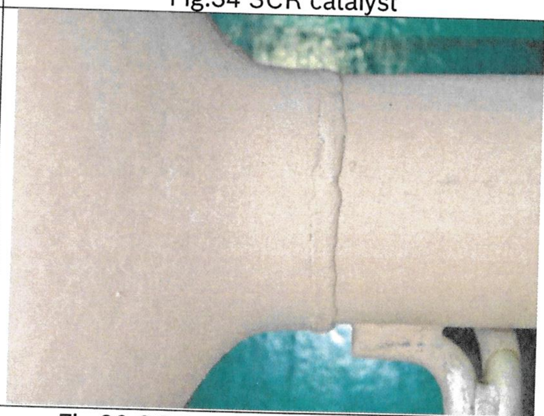

Fig.36 SCR outlet pipe welding seam

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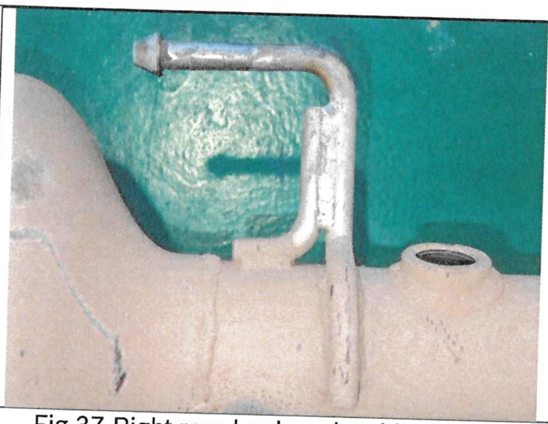

Fig.37 Right rear hook and welding seam

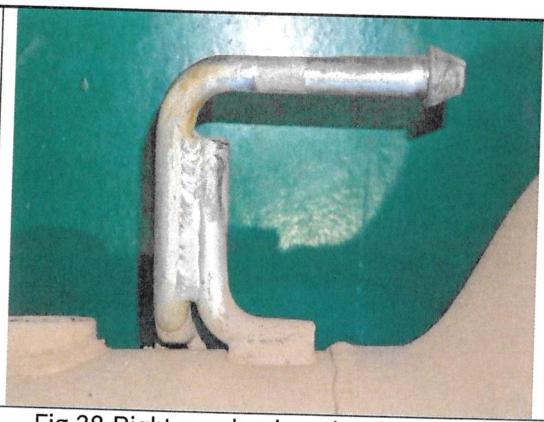

Fig.38 Right rear hook and welding seam

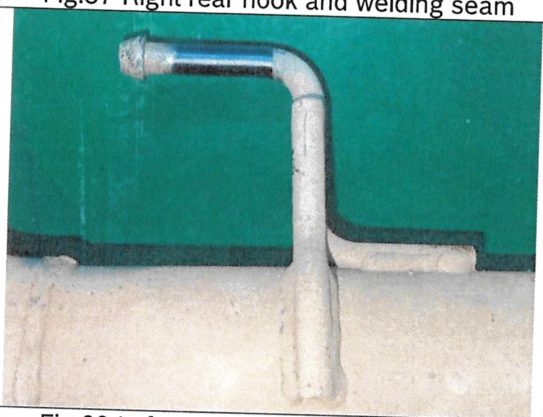

Fig.39 Left rear hook and welding seam

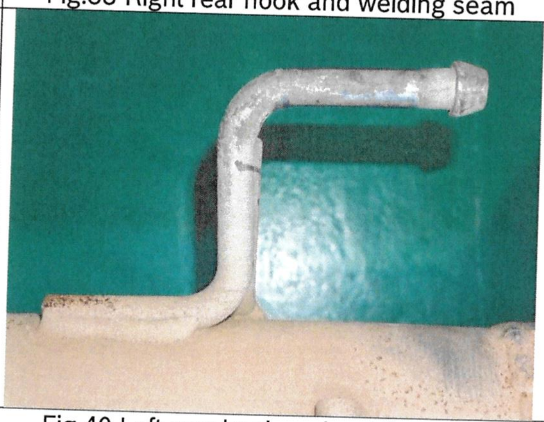

Fig.40 Left rear hook and welding seam

Fig.41 SCR outlet flange welding seam

Fig.42 SCR outlet flange welding seam

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Fig.43 SCR outlet flange